

4 proven ways to dodge costly electronics design mistakes

Through trial, error, and many ‘we should have caught that earlier’ moments, several practices have consistently helped avoid pitfalls in PCB manufacturing:

- **Run regular BoM audits.** Check not only cost but compliance and technical performance.
- **Embrace DFM.** Design for manufacturability reviews catch problems early.
- **Involve both your design partners and manufacturing partners early.** Share your schematics and PCB layout with authorised semiconductor distributors of critical parts that you use (such as STM’s authorised partners), as they often provide design reviews to verify correct component selection, reference designs, power layout, and overall circuit implementation.

Also, share your PCB files with your assembly partner, who can review for manufacturability (DFM), footprint correctness, panelisation, and assembly constraints.

- **Keep supplier communication open.** A quick discussion now can save weeks of troubleshooting later; and source through authorised distributors.
- **Check MSL requirements.** Identify moisture-sensitive components and define proper storage and baking procedures. To prevent failures, such components must be stored in vacuum-sealed packs or dry cabinets, tracked for exposure time, and baked if limits are exceeded, as defined by IPC J-STD-033.
- **Add proper fiducials.** Include global and local fiducials to ensure accurate automated assembly alignment.

In short, catching issues with fiducials and apertures isn’t about luck—it is about building in these habits early.

only boosts order volumes and improves pricing but also simplifies sourcing.

It is worth asking your distributor which parts are in high demand among their other big customers—you might get bulk pricing benefits without committing to massive purchases yourself. Always respect component lifecycles by checking manufacturer websites for end-of-life or NRND status, rather than relying solely on distributor listings. Shelf life varies by component type—electrolytic capacitors, for example, have shorter storage limits and can degrade if kept unused for extended periods.

Watch out for moisture sensitivity levels (MSL). It indicates how vulnerable electronic components are to absorbing moisture before soldering. Components in plastic packages can absorb humidity, and during reflow (above ~240°C), this moisture expands rapidly, causing cracks or the ‘popcorn’ effect.

For example, some LEDs are MSL level 5, giving you just 48 hours to assemble them once opened before they require baking; miss that window, and your production line could come to a halt. So the next

time you are finalising your BoM, remember: treat it not as a static list, but as a living strategy. Your future self—and your budget—will thank you.

A well-optimised BoM can lower costs, increase product reliability, and provide multiple sourcing options, preventing unpleasant last-minute surprises. The challenge is that BoM optimisation is never a one-time task.

Component prices fluctuate, distributors change, and vendors occasionally disappear.

For cost-sensitive products, reviewing the BoM every three to four months is advisable. For lower-volume or less price-sensitive builds, a review once or twice a year is generally sufficient.

Bake, seal, protect

In electronics manufacturing, components may appear robust but are often surprisingly sensitive, particularly when moisture is involved.

MSL is one of those small details that can create serious problems if ignored.

Resistors and capacitors generally tolerate higher moisture exposure, but LEDs, wireless

modules, and integrated circuits are significantly more fragile. When these components remain exposed to air for too long, baking becomes necessary.

This process involves controlled low-temperature heating, sometimes for several days, depending on whether the components are stored in reels or trays. Excessive heat can damage plastic packages, while insufficient heat may fail to remove the trapped moisture.

Some situations are more frustrating than watching an LED ‘pop’ during reflow because trapped moisture expands rapidly inside the package.

Even bare PCBs have moisture sensitivity limits, which makes proper vacuum sealing equally important.

Compliance also requires careful attention. RoHS and lead-free standards are not simply checklist items for individual components. Compliance must extend across the entire manufacturing process.

Solder paste, PCB fabrication, and assembly procedures must all align with the same compliance requirements.

Even a single lead-containing component introduced into an otherwise lead-free assembly can compromise the entire manufacturing process and render the product non-compliant.

DFM and DFA – The overlooked pitfalls

Design for manufacturing (DFM) and design for assembly (DFA) are not merely buzzwords; they often determine whether a project succeeds or fails. Manufacturing is a careful balance of design, materials, and process. It is not enough to focus only on the schematic or the BoM. Every design decision must be evaluated in terms of how it will play out on the assembly floor.

Moisture, heat, weight, and even tiny copper details can determine whether a product moves smoothly